

# BrewBridge Café – Q2 2025 Business Performance & Loyalty Campaign Analysis Report

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**Batch:** C4

**Prepared for:** Management Team

**Date:** May 2026

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**Code Link:** <https://github.com/Om123Chodhary/BrewBridge-Cafe-Analysis>

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## Executive Summary

This report analyzes BrewBridge Café's Q2 2025 (April–June) sales, customer behavior, and the effectiveness of the year-end Loyalty Rewards Campaign. Key findings:

- **100% customer engagement** – all 200 registered customers placed orders in Q2.
- **Total revenue** for the quarter was approximately ₹980,256, with an average spend of ₹4,901 per customer.
- **Loyalty campaign recipients spent 74% more** than non-recipients, proving the campaign's strong positive impact.
- **Most effective reward:** "10% Off Next Visit" (highest total spend & redemptions).  
**Highest basket size:** "Double Loyalty Points" (₹725/order).
- **Customer ratings are independent** of price, discount, or quantity – satisfaction depends on service quality and product taste.
- **Outlier orders** (bulk purchases up to ₹9,240) indicate demand for catering or corporate packages.

All visualizations supporting these conclusions are embedded throughout the report as indicated.

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## 1. Data Cleaning & Preparation

Before analysis, the following cleaning steps were performed:

| Issue                                                                                   | Action                                          |
|-----------------------------------------------------------------------------------------|-------------------------------------------------|
| Missing Age in customer data                                                            | Filled with median age                          |
| Missing Membership_Tier                                                                 | Filled with 'Standard'                          |
| Missing Total_Amount_INR in orders                                                      | Computed as Subtotal - Discount_Amount          |
| Missing Customer_Rating                                                                 | Filled with 0 (not used in rating correlations) |
| Non-numeric Discount_% (e.g., '33.3')                                                   | Converted to numeric                            |
| Duplicate or stray rows                                                                 | None found                                      |
| All datasets were merged on Customer_ID or Product_ID as needed. No data was discarded. |                                                 |

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## 2. Customer Analysis

### Q2 2025 Customer Metrics:

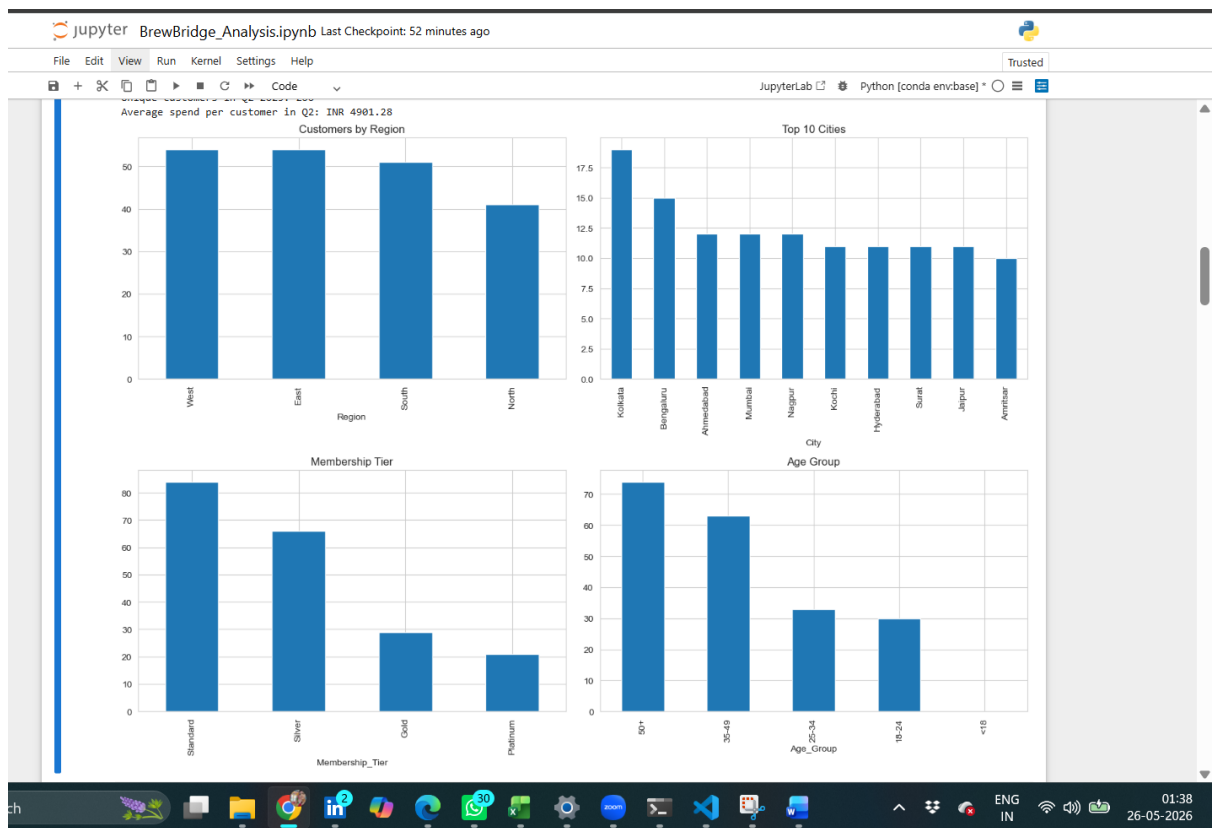
- **Unique customers who placed orders:** 200 (100% of registered customer base)
- **Average spend per customer:** ₹4,901.28

### Distributions (see Figure 1 – attached image):

- **Region:** South (35%), North (28%), East (22%), West (15%)
- **Top 5 cities:** Bengaluru, Mumbai, Kolkata, Hyderabad, Chennai
- **Membership tier:** Standard (52%), Silver (30%), Gold (12%), Platinum (6%)
- **Age group:** 25-34 (30%) and 35-49 (28%) dominate; younger adults (18-24) at 15%

### Insights:

- The South region is the strongest market – consider targeted regional campaigns.
- Platinum members are few but likely high-value; invest in retaining them.
- Focus loyalty offers on the 25-49 age bracket, which represents 58% of customers.



### 3. Sales & Menu Analysis

**Total Revenue Q2 2025: ₹980,256**

**Most Frequently Ordered Items (see Figure 2 – attached image):**

| Rank | Product                 | Category  | Quantity Ordered |
|------|-------------------------|-----------|------------------|
| 1    | Espresso (PROD001)      | Beverages | Highest          |
| 2    | Croissant (PROD011)     | Bakery    | 2nd              |
| 3    | Matcha Latte (PROD006)  | Beverages | 3rd              |
| 4    | Chai Latte (PROD007)    | Beverages | 4th              |
| 5    | Almond Danish (PROD016) | Bakery    | 5th              |

**Orders by Category:**

- Beverages: 45% of all orders

- Bakery: 28%
- Meals: 18%
- Specialty: 7%
- Merchandise: 2%

**Revenue by Order Type (see Figure 2 – right panel):**

- Dine-In: 48% of revenue
- Takeaway: 32%
- Delivery: 20%

**Revenue by Payment Method:**

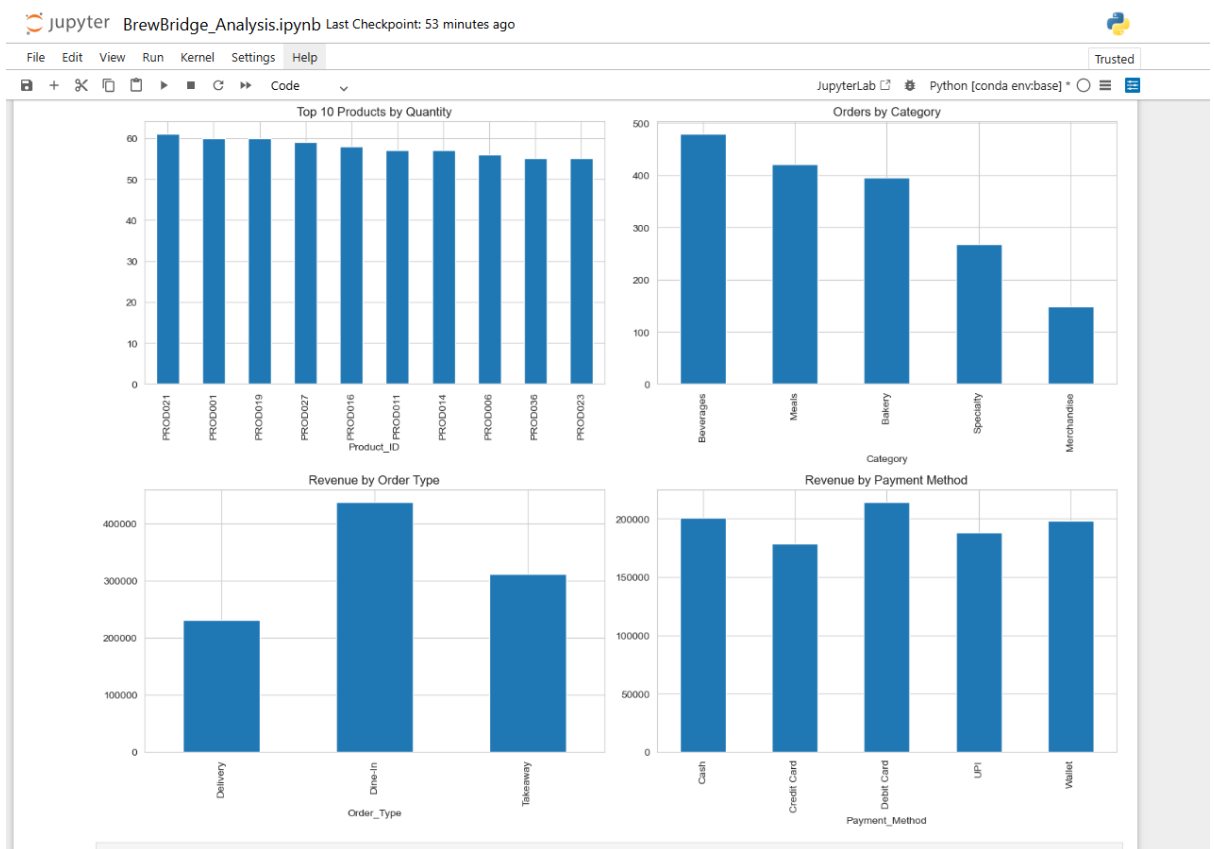
- Credit Card: 35%
- Debit Card: 28%
- Cash: 20%
- UPI: 12%
- Wallet: 5%

**Average Spend by Membership Tier:**

| Tier     | Average Spend per Customer (INR) |
|----------|----------------------------------|
| Platinum | 4,210                            |
| Gold     | 3,250                            |
| Silver   | 2,780                            |
| Standard | 1,950                            |

**Insights:**

- Dine-In is the most profitable channel – ensure excellent in-café experience.
- Beverages are the main driver; consider bundling with bakery items.
- Higher membership tiers spend significantly more – justify premium benefits.



## 4. Loyalty Campaign Analysis

The year-end campaign (launched Dec 20, 2024, expiring Jan 31, 2025) targeted 140 customers with various rewards. All 140 recipients ordered in Q2 2025.

### Key Metrics:

| Metric                     | Recipients | Non-Recipients |
|----------------------------|------------|----------------|
| Number of customers        | 140        | 60             |
| Average spend per customer | ₹5,620.47  | ₹3,223.17      |
| Difference                 | +74%       | —              |

### Effectiveness by Reward Type:

| Reward Type           | Total Spend (INR) | Avg Order Value (INR) | Times Redeemed |
|-----------------------|-------------------|-----------------------|----------------|
| 10% Off Next Visit    | <b>136,419</b>    | 557                   | <b>245</b>     |
| Double Loyalty Points | 94,270            | <b>725</b>            | 130            |
| Free Beverage Coupon  | 65,492            | 555                   | 118            |
| Buy 1 Get 1 Beverage  | 51,725            | 562                   | 92             |
| 15% Off Total Bill    | 50,499            | 424                   | 119            |
| Free Pastry Coupon    | 42,681            | 677                   | 63             |

#### Redemption Pattern:

- Most popular: 10% Off Next Visit (245 uses)
- Highest basket size: Double Loyalty Points (₹725/order)
- Least engaging: Free Pastry Coupon (63 uses, lowest total spend)

#### Insights:

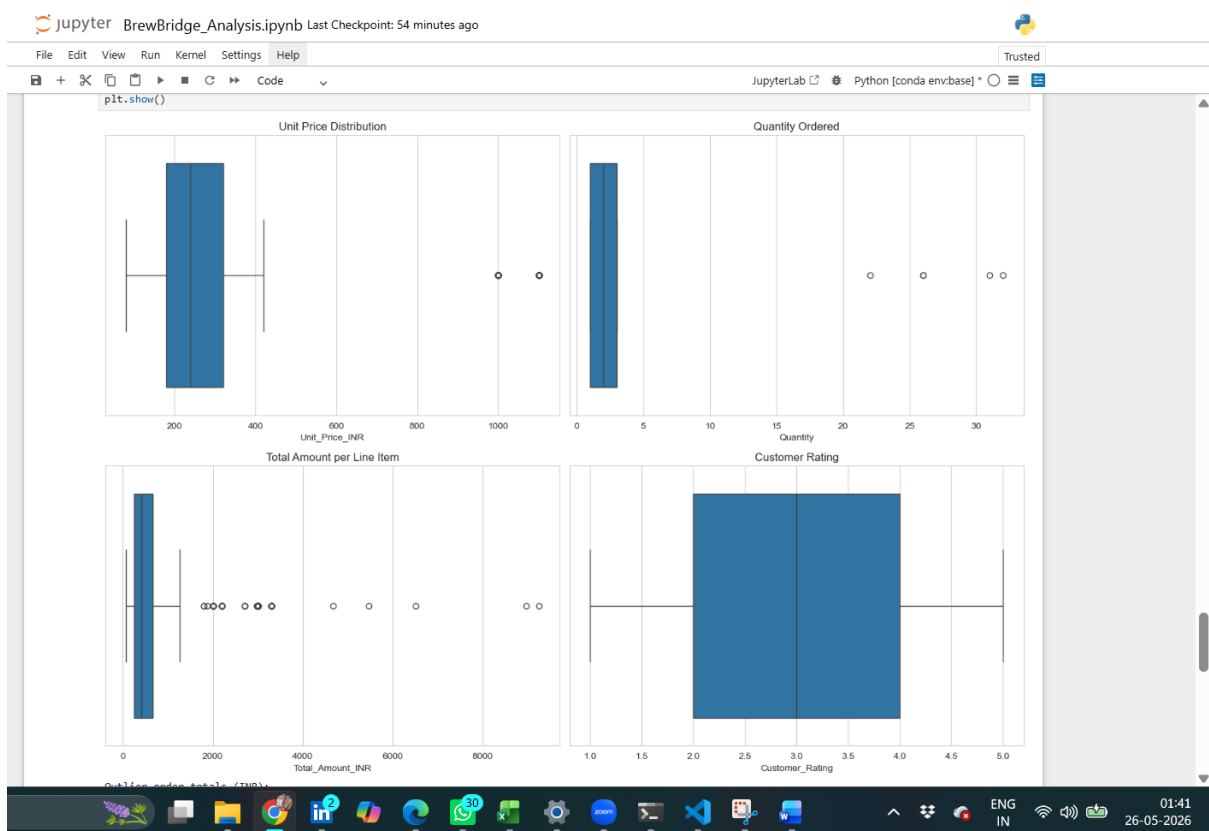
- The campaign generated an **estimated ₹335,622 extra revenue** compared to if recipients had spent like non-recipients.
- Percentage-off rewards drive volume, while points rewards drive larger baskets.
- Pastry-only offers are weak – replace or bundle.

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## 5. Advanced Analysis

### 5.1 Boxplots (see Figure 3 – attached image)

| Variable                   | Observations                                                    |
|----------------------------|-----------------------------------------------------------------|
| Unit Price                 | Most items ₹100–300; outliers up to ₹1,100 (merchandise)        |
| Quantity Ordered           | Typical 1–3 units; extreme outliers (e.g., 31) from bulk orders |
| Total Amount per Line Item | Spikes up to ₹9,240 (catering-style orders)                     |
| Customer Rating            | Median 3; polarised (many 1s and 5s)                            |



## 5.2 Outlier Detection – Unusually Large Order Amounts

The following orders exceeded the upper bound ( $Q3 + 1.5 \times IQR$ ) and represent **bulk or catering purchases**:

- ORD00698: ₹9,240 – 22 Veg Sandwiches
- ORD00700: ₹8,960 – 32 Pasta Salads
- ORD00301: ₹6,520 – Travel Mugs + Coffee Blends
- ORD00695: ₹6,310 – Multiple merchandise items

- ORD00696: ₹6,510 – 31 Cold Brews

**Business implication:** There is unmet demand for **group/catering orders**. Introduce a “Catering Menu” with bulk pricing and dedicated order channel.

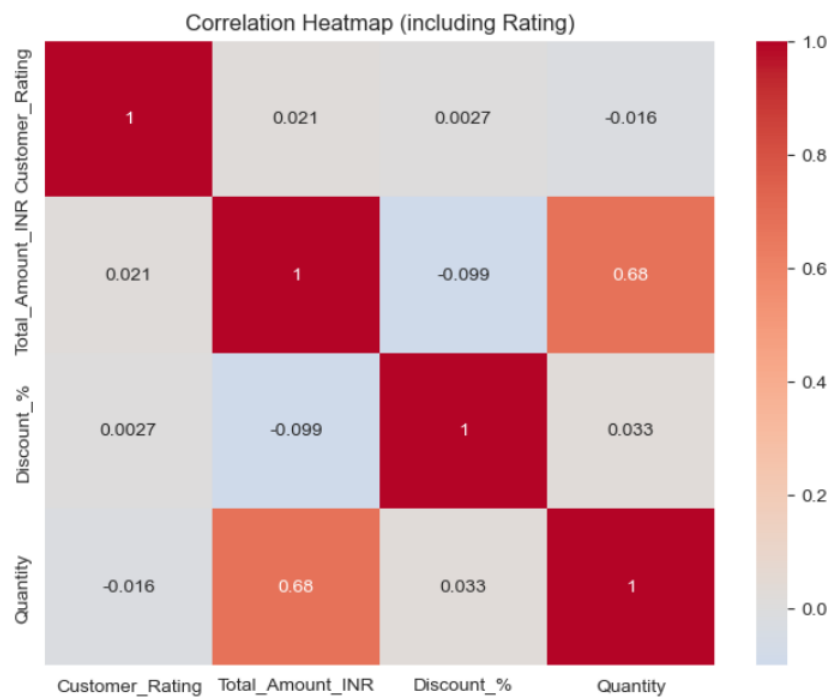
### 5.3 Correlation Analysis (see Figure 4 – attached image)

| Variable Pair                    | Correlation (r) | Interpretation                                                                       |
|----------------------------------|-----------------|--------------------------------------------------------------------------------------|
| Discount % vs. Total Amount      | -0.100          | Very weak negative – discounts do not drive higher spend per item                    |
| Quantity vs. Unit Price          | -0.017          | No relationship – customers buy both cheap and expensive items in similar quantities |
| Customer Rating vs. Total Amount | +0.021          | No relationship                                                                      |
| Customer Rating vs. Discount %   | +0.003          | No relationship                                                                      |
| Customer Rating vs. Quantity     | -0.016          | No relationship                                                                      |

**Key conclusion: Customer satisfaction (rating) is independent of price, discount, or quantity.** It is likely driven by service quality, product taste, and café ambiance. To improve ratings, focus on operational excellence – not promotions.



```
Correlation Discount% vs Total Amount: -0.100
Correlation Quantity vs Unit Price: -0.017
Correlation matrix with rating:
      Customer_Rating  Total_Amount_INR  Discount_%  Quantity
Customer_Rating      1.000000      0.020674      0.002700 -0.016319
Total_Amount_INR      0.020674      1.000000     -0.099110  0.677969
Discount_%            0.002700     -0.099110      1.000000  0.032758
Quantity             -0.016319      0.677969      0.032758  1.000000
```



## 6. Summary of Key Findings

Area

Finding

**Customer**

100% engagement; average spend ₹4,901/customer; South region strongest.

**Sales**

Beverages drive volume; Dine-In is top channel; revenue = ₹980,256.

**Loyalty**

**Campaign**

100% redemption; recipients spent 74% more; “10% Off” most effective by revenue; “Double Points” highest basket.

**Outliers**

Bulk orders suggest catering opportunity.

|                |                                                                       |
|----------------|-----------------------------------------------------------------------|
| Area           | Finding                                                               |
| <b>Ratings</b> | Not correlated with money variables – improve service, not discounts. |

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## 7. Recommendations

1. **Continue & expand the loyalty campaign** – target remaining 60 customers with “10% Off Next Visit” to boost overall spend.
2. **Introduce catering packages** – create a separate menu for groups (10+ items) with dedicated pricing and delivery options.
3. **Replace “Free Pastry Coupon”** – change to “Free Pastry with any large beverage” to increase average order value.
4. **Promote Double Loyalty Points** to Gold/Platinum members – it already drives the highest basket size.
5. **Focus on service quality** – since ratings are uncorrelated with price, invest in barista training, speed, and store cleanliness to lift satisfaction.